

Day 9 - Interview Questions

1. What is a string in Python, and how is it defined?

A string is a sequence of characters enclosed in either single (' ') or double (" ") quotes. For example, "Hello, World!" is a string.

2. Explain the purpose of the len() function when applied to a string.

The len() function is used to determine the length (number of characters) of a string.

3. Explain the split() method in Python and provide an example of its usage.

The split() method is used to split a string into a list of substrings based on a specified delimiter.

4. What is the purpose of the str.strip() method, and how does it work?

The str.strip() method removes leading and trailing whitespace (spaces, tabs, newlines) from a string. It returns a new string with the whitespace removed.

5. How do you convert a string to uppercase and lowercase in Python?

You can use the .upper() method to convert a string to uppercase and the .lower() method to convert it to lowercase.

6. Explain the purpose of the strip(), lstrip(), and rstrip() methods for strings.

strip() removes leading and trailing whitespace from a string.

lstrip() removes leading whitespace from a string.

rstrip() removes trailing whitespace from a string.

7. How can you check if a string contains a specific substring?

You can use the in keyword to check if a substring exists within a string. For example, substring in string.

8. What is the format operator (%) in Python, and what is its primary purpose?

The format operator % in Python is used for string formatting. Its primary purpose is to insert values into placeholders within a string to create formatted strings.

9. Can you use the format operator to format floating-point numbers with a specific number of decimal places? If so, how?

Yes, you can format floating-point numbers with a specific number of decimal places using the % operator and the format specifier %.nf, where n is the number of decimal places you want.

**10. Explain the find() and index() methods for searching substrings in a string.
What is the key difference between them?**

Both find() and index() are used to locate the position of a substring in a string. The key difference is that find() returns -1 if the substring is not found, while index() raises a ValueError.

11. What is the capitalize() method used for, and how does it work?

The capitalize() method capitalizes the first character of a string and converts the rest of the characters to lowercase.

12. Explain the isnumeric(), isalpha(), and isalnum() methods for checking the content of a string.

isnumeric() checks if all characters in a string are numeric.

isalpha() checks if all characters in a string are alphabetic.

isalnum() checks if all characters in a string are alphanumeric (letters or digits).